

Java Inheritance – Multilevel Inheritance Using Interfaces

Objective

To implement multilevel inheritance using interfaces in Java and understand how methods are inherited from one interface to another.

Explanation

- **I2** is the parent interface and declares displayI2().
- **I1 extends I2**, so I1 inherits displayI2() and declares displayI1().
- **C1 implements I1**, so C1 provides implementations for methods from both I1 and I2.
- The main() method creates an object of C1 and calls all three methods.

Correct Complete Program

```
// Parent interface
interface I2 {

    void displayI2();
}

// I1 extends I2.
// This represents inheritance between interfaces.
interface I1 extends I2 {

    void displayI1();
}

// C1 implements I1.
// Because I1 extends I2, C1 must implement methods
// from both I1 and I2.
class C1 implements I1 {

    // Implementation of I2 method
    public void displayI2() {
        System.out.println("Method inherited from I2");
    }

    // Implementation of I1 method
    public void displayI1() {
        System.out.println("Method declared in I1");
    }

    // Method of C1
    void displayC1() {
        System.out.println("Method of Class C1");
    }
}

// Main class
public class Question {

    public static void main(String[] args) {

        // Creating an object of C1
        C1 obj = new C1();

        obj.displayI2();
        obj.displayI1();
        obj.displayC1();
    }
}
```

Step-wise Execution

1. Program execution starts from the main() method.
2. An object obj of class C1 is created.
3. displayI2() is called and its implementation is executed.
4. displayI1() is called and its implementation is executed.
5. displayC1() is called and the method belonging to C1 is executed.
6. The program terminates successfully.

Output

```
Method inherited from I2
Method declared in I1
Method of Class C1
```

Test Case Verification

This program requires no keyboard input. It can be executed multiple times; each run should produce the same output.

Test Case 1

```
Expected Output:
Method inherited from I2
Method declared in I1
Method of Class C1
```

```
Actual Output:
Method inherited from I2
Method declared in I1
Method of Class C1
```

Result: Passed

Test Case 2

```
Expected Output:
Method inherited from I2
Method declared in I1
Method of Class C1
```

```
Actual Output:
Method inherited from I2
Method declared in I1
Method of Class C1
```

Result: Passed

Test Case 3

```
Expected Output:
Method inherited from I2
Method declared in I1
Method of Class C1
```

```
Actual Output:
Method inherited from I2
Method declared in I1
Method of Class C1
```

Result: Passed

Result

The Java program was successfully implemented and executed. It demonstrates multilevel inheritance through interface inheritance, where I1 extends I2 and C1 implements I1. All methods were executed

successfully and the expected output was obtained.